

Dhirubhai Ambani University (DAU)

B.Tech — ICT First Year

Group -47

PC223 – B.Tech Exploration Project

**Smart Energy Monitoring System using
Raspberry Pi and LoRa**

Multi-Node IoT Dashboard for Real-time Electricity Tracking

Student Name	ID No.
Mukku seshank praveen	202501242
Chowturi rahul	202501239
Arya Sonawane	202501243
Aryan Vijaybhai Talsania	202501240
Yaksh Chauhan	202501241

Course	PC223 – B.Tech Exploration Project
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Approx. Budget	₹11000 – ₹13,500
Deadline	13th May 2026

1. Introduction

Electricity distribution in residential and commercial buildings increasingly demands intelligent, real-time monitoring solutions. Conventional energy meters require physical inspection and provide no remote visibility or automated billing. This project addresses that gap by building a Smart Energy Monitoring System — a low-cost, scalable IoT platform that wirelessly collects electricity consumption data from multiple smart meter nodes and presents it through an interactive web dashboard hosted on a Raspberry Pi. Each node is built around a PZEM-004T energy meter module paired with an ESP32 microcontroller and a REYAX RYLR896 LoRa radio. Readings are transmitted wirelessly to a central Raspberry Pi gateway, which stores data in a SQLite database and serves a multi-user Flask web application. Customers log in to view only their own meters, switch between nodes, and see auto-calculated electricity bills.

The system is demonstrated using two active smart meter nodes — each connected to a real 230 V AC load (bulb + holder) — to validate end-to-end data collection, wireless transmission, and dashboard visualisation.

2. Project Description

2.1 Working Principle

The PZEM-004T module measures AC voltage, current, active power, energy (kWh), frequency, and power factor continuously. It communicates with the ESP32 over UART (TTL serial). The ESP32 firmware reads these values at a

configurable interval (default: every 10 seconds), packages them into a compact ASCII payload, and transmits via the RYLR896 LoRa module using AT commands over UART. The Raspberry Pi runs a background Python listener that receives LoRa packets through its own RYLR896 receiver module. Each packet is parsed, tagged with a timestamp, and written to SQLite. The Flask application reads this database and renders the dashboard, accessible to any device on the local network by navigating to the Pi's IP address.

2.2 Key Features

Real-time readings: Voltage (V), current (A), power (W), energy (kWh) per node.

- Multi-user login: Secure session-based auth — each customer sees only their own meters.
- Unlimited nodes per user: Users can register and switch between any number of meters.
- Automatic bill calculation: $\text{kWh consumed} \times \text{tariff rate} = \text{bill in ₹}$.
- Admin panel: Mentor/admin can manage all users, nodes, and tariff settings.
- Usage history & graphs: Daily and monthly bar charts powered by Chart.js.
- Tamper & high-usage alerts: Threshold-based notifications on the dashboard.
- PDF bill export: Customers can download a formatted monthly bill.
- Responsive UI: Works on phone and laptop browser — no app installation needed.

2.3 System Architecture

The architecture has three tiers:

- (1) Edge Nodes — PZEM-004T + ESP32 + RYLR896 transmitter at each meter location.
- (2) Gateway — Raspberry Pi with RYLR896 receiver, Python listener, SQLite database, and Flask server.
- (3) Client Layer — any smartphone or laptop browser connecting over Wi-Fi.

Layer	Component	Role
Edge node	PZEM-004T	Measures AC voltage, current, power, energy (kWh)
Edge node	ESP32 Dev Board	Reads PZEM data; transmits via LoRa AT commands
Edge node	RYLR896 (Tx)	LoRa radio — sends readings wirelessly to Pi
Gateway	RYLR896 (Rx)	LoRa radio — receives all node packets on Pi
Gateway	Raspberry Pi 4/5	Parses packets; stores in SQLite; runs Flask app
Gateway	SQLite Database	Persists all readings, users, and billing data
Client	Flask Web App	Dashboard served to user's phone/laptop browser

2.3 System Architecture

- Python 3 — core language; pre-installed on Raspberry Pi OS.
- Flask — a lightweight web framework; handles routing, login sessions, and API endpoints.
- SQLite — embedded database; zero-configuration, single-file storage on the Pi.
- HTML + CSS + JavaScript (Chart.js) — responsive frontend for all browsers.
- Arduino IDE (ESP32 core) — firmware development for each node microcontroller.

2.5 Challenges and Mitigation

Challenge	Mitigation Strategy
LoRa packet collision (multiple nodes)	Assign unique node IDs; stagger transmission intervals
PZEM UART conflict with LoRa UART	Use ESP32 hardware UART2 for PZEM, UART1 for RYLR896
Raspberry Pi UART busy	Use USB-to-UART adapter for RYLR896 receiver on Pi

Flask multi-user data isolation	Session-based auth with per-user node assignment in DB
Power fluctuation at demo load	Use stable lab power supply; add 100 μ F decoupling cap

3. Hardware Requirements

The bill of materials below covers a 2-node demonstration setup. The Raspberry Pi acts as the single central gateway regardless of node count. Each additional node requires one ESP32, one PZEM-004T, and one RYLR896 transmitter module.

S. No	Component	Specification	Qty	Cost (₹)
1	Raspberry Pi 4 / 5	2 GB+ RAM, built-in Wi-Fi	1	4,500
2	MicroSD Card + Adapter	16 GB Class 10	2	800
3	Pi Power Adapter	5V 3A USB-C	2	100
4	ESP32 Dev Board	ESP32-WROOM-32 / NodeMCU-32S	3	800
5	PZEM-004T Energy Meter	AC 80–260 V, 100 A, UART	2	1400
6	REYAX RYLR896 LoRa Module	433/868/915 MHz, UART AT-cmd	3	1500
7	Wires, PCB & Connectors	Jumper wires, headers, perfboard	Set	600
8	Enclosures / Boxes	ABS project box	3	600
9	Demo Load (Bulb + Holder)	230 V AC, 40–60 W	2 Sets	200

3.1 Component Notes

- RYLR896 quantity = 3: Two transmitters (one per node ESP32); one receiver on Raspberry Pi.
- ESP32 quantity = 3: Two for smart meter nodes; one spare / for gateway serial bridge if needed.
- PZEM-004T: Non-invasive CT clamp included; clamps onto live wire. Supports Modbus RTU over UART.
- Demo Load: A standard 230 V AC bulb in a holder, used to create a measurable electrical load during a live demonstration to the faculty mentor.

4. Estimated Timeline of the Project

The project is planned over 12 weeks in the Autumn 2026 semester. Each phase builds on the previous, with full integration testing and documentation in the final weeks.

5. Budget Summary

Component prices are based on current listings on Indian e-commerce platforms such as Robu.in, Tomson Electronics, and Amazon.in. The lower bound assumes reuse of available lab stock (wires and perfboard); the upper bound includes fresh procurement of all items.

6. Estimated Learning Outcomes

- Hands-on experience with Raspberry Pi setup, Linux terminal, and Python scripting.
- Understanding of UART serial communication and hardware module interfacing (PZEM, LoRa).
- Practical knowledge of LoRa wireless protocol and AT-command-based radio configuration.
- Web application development using Flask, SQLite, HTML/CSS, and JavaScript.
- Multi-user system design: authentication, role separation, and session management.
- Real-world energy measurement — understanding voltage, current, power, and kWh.
- Project management: planning, iterative prototyping, testing, and

documentation.

- Exposure to IoT system design — edge sensing, wireless transmission, gateway architecture.

8.Declaration

All team members hereby declare that the project idea and the details presented in this report are accurate to the best of our knowledge. The project scope, hardware requirements, timeline, and estimated budget have been discussed with and verified by our respective faculty mentor.

We understand that this report will be shared with the lab team to facilitate component procurement for the PC223 course in the Autumn 2026 semester and may be used as a basis for evaluation, backed by the interactions we have had with our faculty mentor